

◆ 1. Comparer

Purpose: Compare **two pieces of data** (requests, responses, payloads, tokens, etc.) to find **differences or similarities**.

📌 Uses:

- **Token comparison:** Check how two session tokens differ (e.g., is the difference only in 1–2 bits? Then it might be predictable).
- **Response diffing:** Send the same request with slightly different inputs and compare responses.
→ Useful for detecting **SQL injection error messages**, **auth bypass**, or **parameter tampering**.
- **Version tracking:** Compare server responses from two different times (before and after login, patching, etc.).
- **Payload testing:** Compare original vs. fuzzed/intruded requests to see exactly what changed.

📌 Modes:

- **Words view:** Highlights differences word by word.
- **Bytes view:** Shows exact byte-level changes (great for encoded tokens).

👉 Example:

Compare two JWT tokens after login attempts. You may notice only the payload differs but the header + signature stay constant — could reveal weak signing.

◆ 2. Decoder

Purpose: Convert (encode/decode) or analyze data in many formats.

📌 Functions:

- **Encoding/decoding:**
 - URL encoding/decoding (%20 → space)
 - Base64 encode/decode
 - Hex / ASCII / Binary
 - HTML/Unicode encoding
- **Hashing:**
 - MD5, SHA1, SHA256, etc. (non-reversible) → for verifying checksums.
- **Smart decode:** Automatically detects format and decodes.
- **Transformation:** Convert data from one representation to another quickly.

📌 Uses:

- Decode **Base64 session tokens** to see if they contain user info (common vuln: storing plaintext usernames/emails).

- Decode/encode **JWTs** to inspect claims.
- URL-decode obfuscated payloads during fuzzing/intrusion.
- Encode payloads in **hex/URL/Base64** before injecting into requests.
- Verify whether “random-looking” token is actually just a timestamp or user ID after decoding.
- Hash a password guess (MD5/SHA1) and compare with a stored hash.

👉 Example:

You intercept a cookie value:

YWRtaW46cGFzc3dvcmQ=

Decoder → Base64 decode → admin:password 🧑🏻 (weak implementation).

🏛️ Quick Difference

Tool	Main Use	Example
Comparer	Find differences/similarities between two values (diff tool)	Compare two server responses to see how an error message changes
Decoder	Encode/decode/transform data between formats	Decode a Base64 session token to reveal hidden info